

40.018 Heuristics and Systems Theory

Project 1 Proposal

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1. Introduction

Bus Route Redesign and Stop Assignment Optimisation

Singapore's public bus network is dense, highly utilised, and affected by time-varying passenger demand. During peak periods, some routes experience overcrowding, while during off-peak periods, buses may run with low occupancy. This project focuses on redesigning bus routes more efficiently to reduce passenger waiting time, overcrowding, and operating costs. For the context of our project, we consider a set of bus stops with known peak-period passenger demand. We split the day into 3 time periods, with them being 1. Morning Peak (7:00-9:00) 2. Off-Peak (13:00-15:00) 3. Evening Peak (18:00-20:00). The objective is to determine which bus stops should be assigned to each bus service, and the sequence in which each bus service should visit its assigned stops.

Each bus service starts and ends at Tampines Bus Interchange (75009). The route must be efficient in terms of total travel distance but should also cover high demand stops. If some stops are not included in any redesigned route, their passenger demand is considered unmet or lost.

2. Problem Background and Formulation

The problem is bi-objective, where we aim to minimise total travel distance and minimise demand loss. It is a variation of the Travelling Salesman Problem because each bus service must determine an efficient visiting sequence of stops.

2.1 Sets and Parameters

Let:

- $N = \{0, 1, 2, \dots, n\}$ be the set of nodes, where node 0 represents the bus interchange, and nodes $1, \dots, n$ represent bus stops
- $S = \{1, 2, \dots, m\}$ be the set of predetermined bus services
- $d_{i,j}$ be the travel distance between stop i and j
- q_i be the peak-period passenger demand at stop i
- K be the maximum number of stops that each bus service can visit
- L_z be the maximum route distance allowed for bus service z
- M be a sufficiently large number used for subtour elimination

where:

- $i, j \in N$ = bus stops or the interchange, $z \in S$ = a bus service

Explanation:

This decision variable represents the number of buses dispatched in time period t . The optimisation algorithm will determine the best values of $x_{i,t}$ to satisfy passenger demand while using available buses efficiently.

2.2 Decision Variables

Route Traversal Indicator	$x_{i,j}^z = \begin{cases} 1, & \text{if bus service } z \text{ travels directly from stop } i \text{ to stop } j \\ 0, & \text{otherwise} \end{cases}$ Indicates whether a bus service travels from stop i to stop j .
Stop Assignment	$y_i^z = \begin{cases} 1, & \text{if stop } i \text{ is assigned to bus service } z \\ 0, & \text{otherwise} \end{cases}$ Indicates whether a bus service visits a particular stop.
Stop Coverage	$u_i = \begin{cases} 1, & \text{if stop } i \text{ is served by at least one bus service} \\ 0, & \text{otherwise} \end{cases}$ Used to calculate demand loss.
Subtour Elimination Variable	$r_i^z = \text{position of stop } i \text{ in the route sequence of bus service } z.$ Prevents disconnected subtours.

2.3.1 Objective 1: Minimise Total Travel Distance

This objective minimises the total distance travelled by all redesigned bus services.

$$\min Z_1 = \sum_{z \in S} \sum_{i \in N} \sum_{j \in N, j \neq i} d_{i,j} x_{i,j}^z$$

2.3.2 Objective 2: Minimise Demand Loss

This objective encourages the model to serve high-demand stops and reduce demand loss.

$$\min Z_2 = \sum_{i \in N \setminus \{0\}} q_i (1 - u_i)$$

where: $u_i = 1$ if stop i is included in at least one route

2.3.3 Bi-Objective

This objective encourages the model to serve high-demand stops and reduce demand loss.

$$\min(Z_1, Z_2)$$

where Z_1 = total travel distance and Z_2 = total demand loss

2.4 Constraints

All services start from the interchange	$\sum_{j \in N, j \neq 0} x_{0,j}^z = 1, \quad \forall z \in S$ Each bus service must leave the interchange ($i=0$) exactly once
All services return to the interchange	$\sum_{i \in N, i \neq 0} x_{i,0}^z = 1, \quad \forall z \in S$ Each bus service must return to the interchange ($j=0$)
Flow Conservation	$\sum_{j \in N, j \neq i} x_{i,j}^z = \sum_{j \in N, j \neq i} x_{j,i}^z, \quad \forall i \in N \setminus \{0\}, z \in S$ If a bus service enters a stop, it must also leave that stop
Linking stop assignment to route traversal	$\sum_{j \in N, j \neq i} x_{i,j}^z = y_i^z, \quad \forall i \in N \setminus \{0\}, z \in S$ If service z visits stop i, then it must leave stop i exactly once $\sum_{j \in N, j \neq i} x_{j,i}^z = y_i^z, \quad \forall i \in N \setminus \{0\}, z \in S$ If service z visits stop i, then it must enter stop i exactly once
Each stop can be served by 1 or more services	$\sum_{z \in S} y_i^z = 0, 1, 2, \dots$ Allows for overlapping of bus services being assigned to a stop
Defining whether a stop is covered	$u_i \leq \sum_{z \in S} y_i^z, \quad \forall i \in N \setminus \{0\}$ $u_i \geq y_i^z, \quad \forall z \in S$ If no service visits stop i, then $u_i = 0$ If at least one service visits stop i, then $u_i = 1$ If multiple services visit stop i, u_i still stays at 1.
Maximum number of stops per service	$\sum_{i \in N \setminus \{0\}} y_i^z \leq K, \quad \forall z \in S$ Prevents a bus service from taking too many stops
Maximum Route Distance per service	$\sum_{i \in N} \sum_{j \in N, j \neq i} d_{i,j} x_{i,j}^z \leq L_z, \quad \forall z \in S$ Prevents route from becoming too long
Subtour Elimination	For all $i, j \in N \setminus \{0\}, i \neq j$, and for each service z: $r_i^z - r_j^z + K x_{i,j}^z \leq K - 1, \quad \forall i \neq j, z \in S$ Prevents the model from creating small disconnected loops that do not include the interchange
Additional Binary Restrictions	$x_{i,j}^z \in \{0,1\}, \quad \forall i, j \in N, i \neq j, z \in S$ $y_i^z \in \{0,1\}, \quad \forall i \in N \setminus \{0\}, z \in S$ $u_i \in \{0,1\}, \quad \forall i \in N \setminus \{0\}$

3. Data and Preprocessing Steps

4 main data sources were used, obtained from LTA Datamall and categorised into 4 components: passenger volume, bus headway time, bus services data and bus stop locations.

- 3.1. Passenger Volume Data (*total_passenger_vol.csv*)
Used to estimate the passenger load on each bus to calculate the overcrowding penalty.
Sourced from dataset: Passenger Volume by Bus Stops.
- 3.2. Bus Headway Time (*buses_that_loop,OD_at_75009.csv*)
Calculated by finding the time between buses arriving at the interchange, using real-time Bus Arrival information for Bus Services at a queried Bus Stop, including Estimated Time of Arrival (ETA), Estimated Location, Load info.
- 3.3. Bus Services Data
Returns detailed service information for all buses currently in operation, including: first stop, last stop, peak/off-peak frequency of dispatch.
- 3.4. Bus Stop Locations (*dist_from_origin.csv, matched_bus_stops_all_services.csv*)
We obtained the locations of the 447 bus stops of services that loop from Tampines Interchange from data.gov.sg, and calculated the distance between bus stops using data from LTA Datamall.

4. Main Metaheuristic Method: Genetic Algorithm

- 4.1. Crossover Operator
- 4.2. Mutation Operator
- 4.3. Repair Operator
- 4.4. Fitness Function

5. Metaheuristic Method: Simulated Annealing

- 5.1. Local Search Operators

6. Performance Benchmark

- 6.1. Solving a small instance in Gurobi
- 6.2.

7. Insights Gained